Hybrid Hand-Actuated Espresso Machine

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Problem Statement

The goal is to produce a double shot of espresso consistently and per espresso industry standards. A single shot of espresso is defined to be 7-9 grams of finely ground coffee beans that is used to produce a 30mL "shot" of espresso and a double shot is 14-18 grams producing a 60mL shot. The extraction process uses 9 bars of pressure, equivalent to approximately 130 psi of pressure that presses water through a tamped "puck" of espresso contained within a 58mm portafilter. A fine mesh shower screen is used to evenly distribute water over the top of the puck during the extraction process. The tamping process uses anywhere from 20-30 lbs of pressure to compress the finely ground coffee beans, which is a process that occurs externally from the espresso machine and will be controlled by a calibrated tamper purchased during the testing phase of the build process. The temperature control of the extraction process is also considered with optimal temperature ranges for a shot of espresso being between 105-205 °Fahrenheit, which is to be controlled in the water heating tank. Due to the unique qualities of each type of coffee bean and the interfacing between the brewing machine, the individual user often has to fine-tune the size of the coffee grinds, the amount of dry coffee, and the exact temperature of the water being pulled through the puck.

Function/Constraints

- Pull a double shot of espresso through 58mm portafilter at 9 bar of pressure
- Water tank is to contain enough water for 5 double shots of espresso, accommodating for approx. 15mL of waste fluid per single shot and water is to be heated to approximately 195-205 F before reaching brew chamber
- Actuation force by user is to be around 5 lbs on lever
- The machine is to be able to pull shots with a variance of ±10% volume in 25-30 seconds from beginning of crank action to end
- Appliance must fit within 2x2x2 ft^3 area

Background

Espresso is highly concentrated form of coffee, made by pushing near boiling water through finely ground coffee in a quick time frame. The fine ground coffee is poured into the 58mm portafilter and then compressed inside the portafilter. A shower screen is placed on top of the grounds and the portafilter is locked into place in the group head on the machine. For a double shot, approximately 90mL of water is dispensed into the brew chamber. The user then uses the hand-crank to depress the piston into the brew chamber and press the hot water through the coffee puck, producing a delicious shot of espresso.

Design

The **Espresso Machine** is designed around a system of linkages combined with a worm gear to generate the substantial force required for extraction. This mechanism produces nearly 500 pounds of force, achieving an overall mechanical advantage of approximately 120:1.

Water heating is accomplished using a coiled copper tube wrapped with high-resistance wire, forming a compact and efficient resistive heating element.

Control and system operation are managed electronically using a Raspberry Pi, with the entire system powered from standard wall power.

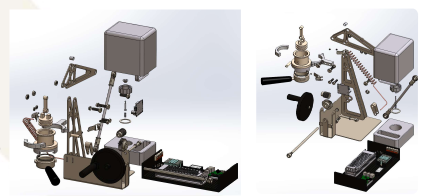


Figure 2: Exploded Views

Figure 7: Close-Up Gear Mesh

Figure 1: Rendering of Full Assembly

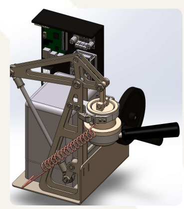


Figure 3: Isometric View

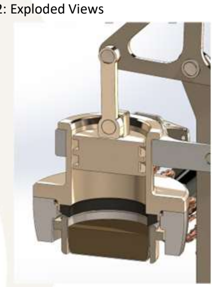


Figure 4: Section View

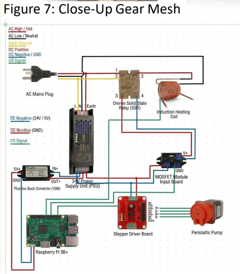


Figure 5: Full Electrical Diagram

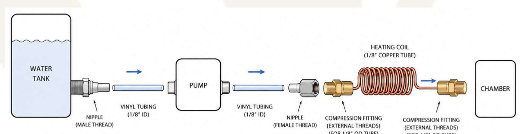
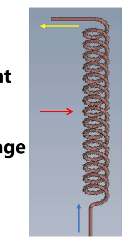


Figure 6: Full Plumbing Diagram

Analysis

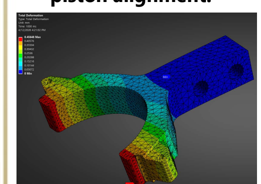
Water Heating: Finding Power requirement

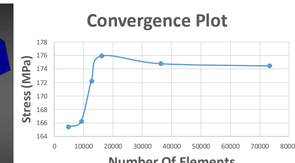
- **Energy required: Q=m*C_p*ΔT**
- **Wattage needed: W=Q/t**
- **Length of wire required to supply wattage**
- **Length of pipe required using constant surface heat flux condition**



Ansys FEA: Deflection of Clamp

- **Minimizing deflection of brew chamber to maintain piston alignment.**





Testing

Next semester we plan to testing the following parameters:

- **Water temperature and volumetric flow.**
- **Liquid retained in coffee grind puck**
- **Brew chamber deflection under load**
- **Actuation force**
- **Pressure consistency throughout shot**